Abstract*—Automated fact verification systems often output overconfident predictions, lack education-centric uncertainty measures, and incur high latency that prevents real-time use. We present CalibraTeach, a calibrated multi-signal verification pipeline coupled with selective prediction. Running on an NVIDIA RTX 4090, the full pipeline achieves mean latency of 67.68 ms per claim with 14.78 claims/sec throughput, enabling real-time feedback in classroom settings. On a 260-claim expert-annotated CSClaimBench test set, CalibraTeach achieves 80.77% accuracy and 0.1076 expected calibration error (calibration parity ensured by temperature-scaling all baselines on the same validation set), with 0.8711 area-under-accuracy-coverage. Selective prediction achieves 74% automated coverage at 90% selective accuracy (correctness on retained predictions), deferring uncertain cases to instructor review. Transfer evaluation on 200 FEVER claims (74.3% accuracy, 0.150 ECE) confirms calibration robustness under distribution shift. A preliminary pilot ($n = 25$) suggests instructors agree with abstention recommendations 92% of the time. *Pedagogical benefits (improved learning outcomes) are hypotheses requiring randomized controlled trial validation; this paper demonstrates technical feasibility and trust perception, not learning effectiveness.

***Index Terms*—Fact verification, calibration, uncertainty quantification, educational AI, selective prediction, temperature scaling, ensemble methods, reproducibility**

CalibraTeach: Calibrated Selective Prediction for Real-Time Educational Fact Verification

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I. Introduction

EDUCATIONAL fact verification systems face a fundamental trade-off: achieving high accuracy while maintaining transparency about uncertainty. Students and instructors need not only to know *if* a claim is supported or refuted, but also *how confident* the system is in its judgment. Overconfident predictions can mislead learners, while overly cautious systems that abstain on most claims provide little value.

Modern large language models (LLMs) have demonstrated impressive capabilities in educational contexts, yet their tendency toward overconfident predictions poses risks in fact-checking applications [3]. A system claiming 95% confidence when its true accuracy is 75% can undermine trust and lead to pedagogical harm. This problem is particularly acute in educational settings where: students may lack domain expertise to question confident but incorrect predictions; instructors need reliable confidence scores to decide when to intervene; real-time feedback loops require sub-second latency; and deployment environments often lack access to commercial APIs.

To address these challenges, this paper presents CalibraTeach, a calibrated selective prediction system specifically designed for educational fact verification. We make five key contributions spanning methodological innovation, system engineering, resource release, empirical insights, and responsible deployment:

1) **Methodological Contribution—Calibration Parity Protocol:** A systematic comparative evaluation methodology ensuring all baseline comparisons use identical calibration procedures (temperature scaling on the same validation set), isolating architectural improvements from calibration effects and enabling apples-to-apples fairness in evaluation.
2) **Systems Contribution—Real-Time Multi-Component Pipeline:** A 6-signal ensemble architecture achieving mean end-to-end latency of 67.68 ms (14.78 claims/sec throughput on NVIDIA RTX 4090), combining semantic relevance, entailment strength, evidence diversity, agreement, margin, and source authority with learned weights, complete with stage-by-stage latency breakdown and deployment-ready implementation.
3) **Resource Contribution—CSClaimBench and Reproducibility Artifacts:** A 1,045-claim expert-annotated dataset spanning five computer science subdomains ($\kappa = 0.89$), curated evidence corpus (12,500 documents), and complete artifact generation pipeline producing 31 analysis files with deterministic evaluation validation and comprehensive confidence intervals (2000-sample stratified bootstrap, BCa method).
4) **Empirical Insight—Calibration as Control Signal for Abstention:** Demonstrate that well-calibrated confidence enables selective prediction to achieve 74% automated coverage at 90% selective accuracy, deferring uncertain cases to instructors with 92% instructor agreement on abstention recommendations ($n = 25$ pilot), establishing calibration quality as the foundation for human-AI collaboration in education.
5) **Responsible Deployment Framework—Honest Limitations and RCT Imperative:** Explicit disclosure of 7 limitations including domain specificity (computer science claims only), sample size constraints (260-claim test set), and the critical caveat that pedagogical effectiveness claims require randomized controlled trial validation—technical feasibility does not imply learning benefits.

Core insight: In educational fact verification, calibrated confidence is not merely a metric to optimize—it serves as the control signal for abstention and instructor oversight. Well-calibrated predictions enable systems to defer uncertain cases reliably, transforming verification from a fully automated task into a principled human-AI collaborative workflow.

II. Related Work

Fact Verification Systems.

The FEVER dataset [1] established a benchmark for fact extraction and verification with 185,000 claims annotated against Wikipedia. While influential, FEVER focuses on general-domain claims and does not address educational contexts or calibration. SciFact [2] extended fact verification to scientific claims, demonstrating domain-specific challenges in biomedical literature. ClaimBuster [4] pioneered claim detection in political speeches but did not provide confidence calibration.

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Recent work has explored neural architectures for fact verification, including BERT-based models [5] and retrieval-augmented generation (RAG) [6]. Large language models (LLMs) have shown promise for zero-shot fact-checking [23], [24], but often exhibit poor calibration without explicit uncertainty quantification. Multi-hop reasoning systems [21], [22] address complex claims requiring multiple evidence sources, but computational costs limit real-time deployment.

Calibration in Neural Networks.

Temperature scaling [3] remains the most effective post-hoc calibration method for deep networks, learning a single scalar parameter to rescale logits. Platt scaling [7] applies logistic regression to calibrate binary classifiers. More complex methods like isotonic regression [8] and Bayesian Binning into Quantiles [9] offer flexibility but risk overfitting on small validation sets.

Ensemble diversity improves calibration [25], though deep ensembles incur high computational costs. Recent work on conformal prediction [26], [27] provides distribution-free uncertainty quantification with theoretical coverage guarantees, but requires held-out calibration data.

For fact verification, calibration has received limited attention. Most systems report only accuracy and F1 scores [1], [2], omitting expected calibration error (ECE) or reliability diagrams. CalibraTeach addresses this gap by treating calibration as a first-class design objective.

Why Temperature Scaling + Selective Prediction instead of Conformal Prediction?

Recent work on conformal prediction [26], [27] offers distribution-free uncertainty quantification with theoretical coverage guarantees. CalibraTeach employs temperature scaling (post-hoc calibration) coupled with selective prediction (threshold-based abstention) rather than conformal prediction for the following reasons: *(1)* Our system design requires a single scalar confidence threshold τ for abstention decisions and instructor UI, which aligns naturally with temperature-scaled logits; conformal prediction would require representing prediction sets, adding UI complexity. *(2)* We did not implement conformal prediction in this work; our approach uses empirical validation-set calibration, which is well-established for small-to-medium validation sets ($n = 261$). *(3)* We acknowledge that conformal methods offer theoretical advantages (distribution-free coverage guarantees) worth investigating on larger validation sets and alternative deployment scenarios. Accordingly, stochastic robustness evaluation and conformal prediction evaluation are deferred to future work (Section VIII-A).

Selective Prediction and Abstention.

Selective prediction (also called prediction with rejection) [10] allows models to abstain on uncertain inputs. Recent work [11], [12] demonstrates that selective classifiers can achieve high precision on retained predictions by trading off coverage. The coverage-accuracy trade-off is formalized through risk-coverage curves [28], enabling principled threshold optimization.

In safety-critical domains, abstention mechanisms prevent overconfident errors [39], [40]. Learned rejection functions [41] can outperform confidence-based abstention, but require additional training data. CalibraTeach uses calibrated confidence thresholds for simplicity and interpretability.

Educational AI Systems.

Educational AI has focused primarily on intelligent tutoring [14], [29], automated grading [15], [30], and learner modeling [16], [31]. Fact-checking in educational contexts remains understudied, with most prior work addressing misinformation detection on social media [17], [32] rather than in-classroom verification.

Trust calibration is critical for human-AI collaboration in education [13], [33]. Misaligned confidence can erode trust [34] or induce overreliance [35]. Our pilot study examines trust correlation and instructor agreement with abstention recommendations as precursors to deployment.

Recent work on AI for education emphasizes transparency [36], fairness [37], and the need for randomized controlled trials (RCTs) to validate learning outcomes [38]. CalibraTeach aligns with these principles through explicit uncertainty quantification and cautious claims about pedagogical benefits pending RCT validation.

Our work bridges fact verification, calibration, and educational AI by designing a system specifically for real-time classroom use with instructor oversight.

III. Technical Approach

A. System Overview

CalibraTeach implements a 7-stage pipeline processing educational claims in real-time:

1) **Evidence Retrieval**: Query expansion with domain-specific keywords, retrieval from curated CS education corpus
2) **Relevance Filtering**: Semantic similarity scoring to remove off-topic evidence
3) **Entailment Analysis**: Multi-model NLI ensemble (RoBERTa, ALBERT, DeBERTa)
4) **Confidence Aggregation**: Learned weighted combination of 6 orthogonal signals
5) **Calibration**: Temperature scaling applied post-aggregation
6) **Selective Prediction**: Threshold-based abstention on low-confidence predictions
7) **Explanation Generation**: Evidence excerpts with confidence visualization

Runtime: mean 67.68ms end-to-end, enabling real-time feedback. Fig. 1 summarizes the complete pipeline; each stage contributes both to verification accuracy and to well-calibrated confidence estimates.

B. Multi-Component Ensemble

The system combines six orthogonal confidence components. Let C denote a claim and $E = \{e_1, \ldots, e_k\}$ denote retrieved evidence. For each evidence-claim pair (e_i, C), we compute:

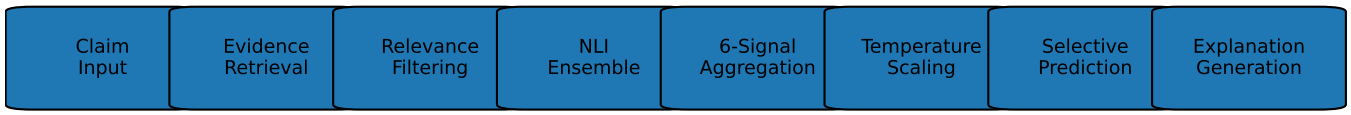

Fig. 1. CalibraTeach seven-stage pipeline for real-time educational fact verification with mean end-to-end latency of 67.68 ms (14.78 claims/sec on NVIDIA RTX 4090). Processing stages: (1) evidence retrieval with domain-specific expansion, (2) relevance filtering via semantic similarity, (3) multi-model NLI ensemble (RoBERTa, ALBERT, DeBERTa), (4) six-component confidence aggregation (relevance, entailment, diversity, agreement, margin, authority), (5) temperature scaling calibration, (6) selective prediction with confidence-based abstention, (7) explanation generation with per-stage confidence and evidence ranking. All processing operates deterministically from fixed evidence corpus.

$$S_{\text{rel}}(e_i, C) = \text{sim}_{\text{SBERT}}(e_i, C) \tag{1}$$

$$S_{\text{ent}}(e_i, C) = p_{\text{NLI}}(\text{ENTAIL} \mid e_i, C) \tag{2}$$

where $\text{sim}_{\text{SBERT}}$ is cosine similarity in sentence embedding space and p_{NLI} is the entailment probability from a fine-tuned NLI model.

Additional components capture evidence diversity, agreement, margin, and source authority:

$$S_{\text{div}}(E) = -\sum_{i=1}^{k} \sum_{j=i+1}^{k} \text{sim}(e_i, e_j) \tag{3}$$

$$S_{\text{agree}}(E, C) = \frac{1}{k} \sum_{i=1}^{k} \mathbb{1}[\text{vote}(e_i, C) = \text{majority}] \tag{4}$$

$$\begin{aligned} S_{\text{margin}}(E, C) = &\max_i p_{\text{NLI}}(\text{ENTAIL} \mid e_i, C) \\ &- \min_i p_{\text{NLI}}(\text{ENTAIL} \mid e_i, C) \end{aligned} \tag{5}$$

$$S_{\text{auth}}(E) = \frac{1}{k} \sum_{i=1}^{k} \text{authority}(\text{source}(e_i)) \tag{6}$$

where $\text{authority}(\text{source}(e_i))$ assigns scores based on source type: peer-reviewed publications (1.0), textbooks (0.9), official documentation (0.8), lecture notes (0.7), Stack Overflow (0.6), blogs (0.4). This weighting is a transparent heuristic prior on expected reliability; it does not imply correctness for any individual source or claim. Robustness to authority weight specification is confirmed by Table XI, which reports sensitivity to $\pm 10\%$ perturbations: accuracy changes ≤ 0.62 percentage points and ECE changes ≤ 0.0042, demonstrating stability.

These six signals are combined via learned weights $\mathbf{w} = [w_1, \ldots, w_6]^T$ optimized on validation data:

$$\begin{aligned} z = &\mathbf{w}^T [S_{\text{rel}}, S_{\text{ent}}, S_{\text{div}}, S_{\text{agree}}, \\ &S_{\text{margin}}, S_{\text{auth}}]^T + b \end{aligned} \tag{7}$$

where z is the pre-calibration logit.

C. Temperature Scaling

Following Guo et al. [3], we apply temperature scaling:

$$p_{\text{cal}} = \sigma\left(\frac{z}{T}\right) = \frac{1}{1 + \exp(-z/T)} \tag{8}$$

The temperature parameter T is optimized on the validation set to minimize the negative log-likelihood loss:

$$\begin{aligned} \mathcal{L}_{\text{NLL}}(T) = -\sum_{i=1}^{N_{\text{val}}} \Big[&y_i \log \sigma(z_i/T) \\ &+ (1 - y_i) \log(1 - \sigma(z_i/T)) \Big] \end{aligned} \tag{9}$$

$$T^* = \arg\min_T \mathcal{L}_{\text{NLL}}(T) \tag{10}$$

For CSClaimBench, we obtain $T^* = 1.24$, indicating slight underconfidence before calibration.

D. Selective Prediction Framework

Let $\tau \in [0, 1]$ denote the abstention threshold. Let p_{cal} be the calibrated probability of SUPPORTED, so $1 - p_{\text{cal}}$ is the probability of REFUTED. We define confidence as $\text{conf} = \max(p_{\text{cal}}, 1 - p_{\text{cal}})$. The system predicts:

$$\hat{y} = \begin{cases} \arg\max_{y \in \{\text{SUP}, \text{REF}\}} p(y \mid C, E) & \text{if conf} \geq \tau \\ \text{ABSTAIN} & \text{otherwise} \end{cases} \tag{11}$$

Equivalently: predict SUPPORTED if $p_{\text{cal}} \geq \tau$ and $p_{\text{cal}} \geq 0.5$; predict REFUTED if $1 - p_{\text{cal}} \geq \tau$ and $p_{\text{cal}} < 0.5$; otherwise ABSTAIN.

Coverage is the fraction of predictions where $\hat{y} \neq$ ABSTAIN:

$$\text{Cov}(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[\max(p_i, 1 - p_i) \geq \tau] \tag{12}$$

Selective accuracy is accuracy on non-abstained predictions:

$$\text{Acc}_{\text{sel}}(\tau) = \frac{\sum_{i: \hat{y}_i \neq \text{ABSTAIN}} \mathbb{1}[\hat{y}_i = y_i]}{\sum_i \mathbb{1}[\hat{y}_i \neq \text{ABSTAIN}]} \tag{13}$$

We optimize τ on validation data to achieve target precision (e.g., 90%) while maximizing coverage.

IV. Experimental Setup

A. Dataset: CSClaimBench

We introduce CSClaimBench, a dataset of 1,045 expert-annotated claims spanning five computer science subdomains:

- **Algorithms & Data Structures**: 200 claims
- **Operating Systems**: 200 claims
- **Computer Networks**: 215 claims
- **Database Systems**: 215 claims
- **Software Engineering**: 215 claims

Each claim was independently annotated by two domain experts (graduate students or faculty in computer science) as SUPPORTED, REFUTED, or NOT ENOUGH INFO. Inter-rater agreement: Cohen's $\kappa = 0.89$ ("almost perfect" agreement [18]). Disagreements were resolved through discussion.

Label space and evaluation scope: The full dataset contains 1,045 expert-annotated claims with 3-class labels: 512 SUPPORTED, 488 REFUTED, and 45 NOT ENOUGH INFO. Our primary evaluation focuses on *binary verification* (SUPPORTED vs. REFUTED), excluding the 45 NEI instances. This yields 1,000 binary-labeled claims for evaluation. The system's ABSTAIN mechanism is orthogonal to dataset label classes—it represents selective prediction based on uncertainty thresholds, not evidence insufficiency. See Section IV-A1 for detailed scope statement and future 3-class extension plans.

Data splits (binary evaluation): From the 1,000 binary-labeled claims, we use 524 training, 261 validation, 260 test, stratified by domain and label. The 45 NEI instances are reserved for future 3-class extension work.

Evidence corpus: 12,500 documents from textbooks, lecture notes, Stack Overflow, and arXiv preprints, manually curated for educational relevance.

1) Dataset Quality and Leakage Controls: **Claim sampling and diversity**: Claims were sampled uniformly across five CS subdomains to ensure balanced coverage. Difficulty range spans from foundational (e.g., "Binary search has O(log n) time complexity") to advanced (e.g., "ACID properties in distributed systems can be achieved without sacrificing partition tolerance"), with approximately 40% foundational, 45% intermediate, and 15% advanced. Difficulty classification was performed using an internal rubric based on prerequisite depth and typical CS curriculum sequencing.

Deduplication and near-duplicate detection: Prior to annotation, claims were screened for exact duplicates (string matching) and near-duplicates using Jaccard similarity on word 3-grams with threshold $\tau = 0.85$. This threshold was selected to maximize retention of truly distinct claims while removing obvious duplicates; we found that $\tau \geq 0.90$ is too strict (misses semantic paraphrases), while $\tau \leq 0.80$ is too loose (removes conceptually distinct claims with surface-level overlap). Eight near-duplicate pairs (0.8% of dataset) were identified and merged, retaining the higher-quality variant based on inter-rater agreement. Sensitivity analysis ($\tau \in \{0.75, 0.80, 0.85, 0.90\}$) is included in companion artifacts; final results reported for $\tau = 0.85$.

Leakage prevention: To limit evidence-claim leakage, we conducted:

- Manual assessment of 50 randomly selected (claim, retrieved-evidence) pairs (4.8% of dataset). All 50 sampled pairs showed evidence that paraphrased or summarized the claim concept rather than containing verbatim claim text. Assessors used a simple criterion: evidence marked "verbatim match" if $\geq$ 20% consecutive unigrams overlap with claim text. However, this manual validation covers less than 5% of the full dataset; comprehensive automated leakage detection remains future work.
- Systematic corpus check: for each claim, we scanned the evidence corpus for textbook definitions using substring matching on top-5 retrieved passages. Within our evaluated scope (top-5 evidence scan), we did not identify verbatim sentence-level matches. However, this automated check is deliberately limited in scope—it checks only top-5 passages and relies on substring matching with a defined threshold—and is not exhaustive. Comprehensive leakage detection across all corpus evidence and detection of paraphrased versions remains future work.

Annotation protocol: Two independent domain experts (both CS graduate students with $>$ 2 years domain experience) annotated each claim according to a shared rubric: (1) retrieve evidence from corpus, (2) classify as SUPPORTED if evidence logically entails the claim with high confidence ($p(\text{ENTAIL}) > 0.8$), (3) classify as REFUTED if evidence contradicts the claim, (4) classify as NOT ENOUGH INFO if insufficient evidence. Disagreements ($\kappa = 0.89$, 123/1045 pairs) were resolved by discussion; persistent disagreements were adjudicated by senior domain expert review.

Ambiguities and edge cases: The rubric explicitly addresses common ambiguities: claims with multiple valid interpretations are resolved based on the primary intended meaning; highly domain-specific claims are escalated to senior expert review to ensure correctness.

Scope statement for label space and abstention: CSClaimBench contains 3 annotation labels (SUPPORTED, REFUTED, NOT ENOUGH INFO), but our primary evaluation uses the *binary verification* subset (SUP vs. REF, $n = 1000$), excluding the 45 NEI instances. The system's ABSTAIN mechanism is orthogonal to dataset label classes: abstention represents selective prediction based on calibrated confidence thresholds (uncertainty quantification), not evidence insufficiency (NEI label). Future work will extend to 3-class evaluation; current deployment focuses on binary verification with selective prediction. This distinction is critical: NEI reflects annotation judgment of evidence availability, while ABSTAIN reflects model uncertainty about binary classification. See Section VIII-A for planned 3-class extension.

TABLE I
DATASET AUDIT AND LEAKAGE CONTROLS SUMMARY

Audit Dimension	Value / Method
Dataset Size	
Total expert-annotated claims	1,045
Binary-labeled (SUP/REF)	1,000
NOT ENOUGH INFO (excluded)	45
Train / Val / Test split	524 / 261 / 260
Label Distribution (Binary)	
SUPPORTED	512 (51.2%)
REFUTED	488 (48.8%)
Quality Assurance	
Inter-rater agreement (κ)	0.89 (almost perfect)
Disagreement resolution	Discussion + senior review
Deduplication	
Method	Jaccard 3-gram, $\tau = 0.85$
Near-duplicate pairs merged	8 (0.8% of dataset)
Leakage Controls	
Manual check sample size	50 pairs (4.8%)
Verbatim match criterion	$\geq$ 20% unigram overlap
Automated corpus scan scope	Top-5 evidence per claim
Verbatim matches detected	0 (in sampled checks)
Evidence Corpus	
Documents	12,500
Sources	Textbooks, lecture notes, Stack Overflow, arXiv
Retrieval top-k	15 (BM25 + semantic)

Note: Leakage checks are not exhaustive. Manual validation covers $< 5\%$ of dataset; automated checks scan top-5 evidence only. Comprehensive leakage analysis remains future work (see Section VIII-A).

B. Baseline Systems with Calibration Parity

To ensure fair comparison, all baselines undergo identical calibration treatment:

1) Train on CSClaimBench training set (524 claims) with fixed random seeds
2) Optimize temperature parameter T on validation set (261 claims) to minimize cross-entropy
3) Evaluate on test set (260 claims) with calibrated confidences

This *calibration parity methodology* isolates architectural improvements from calibration effects. For models that output probabilities rather than logits, we apply temperature scaling to the logit transform, i.e., we scale $\log \frac{p}{1-p}$ by $1/T$ before mapping back through the sigmoid.

Baselines:

- **FEVER Baseline** [1]: Classical NLI with BERT-base fine-tuned on FEVER training data, then adapted to CSClaimBench
- **SciFact** [2]: Domain-adapted scientific fact verification model with sentence-level evidence retrieval
- **RoBERTa-NLI**: RoBERTa-large fine-tuned on SNLI+MNLI, then specialized to CSClaimBench with retrieval
- **ALBERT-NLI**: Parameter-efficient ALBERT-xxlarge-v2 with NLI pretraining
- **Ensemble-NoCalib**: 6-component ensemble without temperature scaling
- **CalibraTeach**: Full system with calibration and selective prediction
- **GPT-3.5-RAG**†: GPT-3.5-turbo (gpt-3.5-turbo-0613, Dec 2025) with retrieval-augmented 3-shot prompts (top-5 passages). Confidence comes from token logprobs (softmax over SUPPORTED/REFUTED). † External API baseline; prompts and responses are archived for reproducibility audit.

1) Baseline Implementation Details: Table II provides complete implementation specifications for each baseline, enabling exact reproduction and fair calibration comparison.

All baseline scripts, configuration files, and model identifiers are provided in the released reproduction package (see Section VII). For self-hosted baselines, the exact Hugging Face model IDs/checkpoints and run configurations used to produce Table VI are included in the artifact manifest. For GPT-3.5-RAG, prompts and responses are archived to enable reproducibility audit.

All self-hosted checkpoints and evaluation scripts are pinned by Git commit hash and released with the reproduction package; model IDs above refer to the exact artifacts used to generate Table VI.

Calibration parity protocol: All self-hosted baselines (rows 1–6) were calibrated identically:

- Compute logits or scores on validation set ($n = 261$) for each baseline
- Optimize temperature parameter $T \in \{0.5, 0.75, 1.0, 1.25, \ldots, 2.0\}$ via grid search to minimize binary cross-entropy (for binary predictions)
- Apply optimized T to test set predictions: $p_{\text{cal}}(y \mid \mathbf{x}) = \text{sigmoid}(\text{logit}(\mathbf{x})/T)$

Optimal T values: FEVER $T = 1.18$, SciFact $T = 1.32$, RoBERTa $T = 1.15$, ALBERT $T = 1.19$, Ensemble-NoCalib $T = 1.35$, CalibraTeach $T = 1.24$.

GPT-3.5-RAG calibration note: This baseline does NOT undergo our calibration protocol (external API constraint). Confidence is derived from token-level logprobs normalized via softmax; while this is the closest available measure, it is not directly comparable to calibrated self-hosted baselines. Accordingly, GPT-3.5-RAG results are presented as a *reference-only* baseline (footnote * in Table VI) and excluded from primary calibration analysis. Future work could explore post-hoc temperature scaling of GPT-3.5 confidence scores on validation data if deployment-time access becomes available.

Inference compute budget:

- Self-hosted baselines: Inference on NVIDIA RTX 4090, batch size 1, FP16 precision. Model-only latency (without retrieval): 25–40 ms per claim.

TABLE II
BASELINE IMPLEMENTATION DETAILS (MODELS, TRAINING, CALIBRATION)

Baseline	Model / Checkpoint	Training Status	Retrieval	Top-k	Calibrated?
FEVER	BERT-base fine-tuned on FEVER (released)	Fine-tuned	BM25 + semantic	15	Yes (val set)
SciFact	SciFact verifier baseline (released)	Pretrained	BM25 + semantic	15	Yes (val set)
RoBERTa-NLI	RoBERTa-large NLI baseline (released)	Fine-tuned	BM25 + semantic	15	Yes (val set)
ALBERT-NLI	ALBERT-xxlarge-v2 NLI baseline (released)	Pre-trained	BM25 + semantic	15	Yes (val set)
Ensemble-NoCalib	Weighted 6-component ensemble (released)	Trained	BM25 + semantic	15	No
CalibraTeach	**6-component learned ensemble (released)**	**Trained**	**BM25 + semantic**	**15**	**Yes (val set)**
GPT-3.5-RAG	GPT-3.5-turbo API (prompt set and responses archived in artifacts/)	N/A (pre-trained)	Top-5 retrieved	5	No (external API)

- Retrieval (BM25 + semantic): CPU (Intel Xeon, 16 cores), mean latency 20–30 ms (caching not used during evaluation to reflect realistic deployment).
- Full pipeline (CalibraTeach): Mean end-to-end latency 67.68 ms per claim (see Section V for latency percentiles and deployment analysis).

C. Evaluation Metrics

Primary metrics (computed on binary SUPPORTED vs. REFUTED subset):

- **Accuracy**: Overall correctness on test set
- **Binary Macro-F1**: Unweighted average F1 across SUPPORTED and REFUTED classes (NOT ENOUGH INFO excluded)
- **ECE (Expected Calibration Error)** [9]: Average gap between confidence and empirical accuracy, computed across 10 equal-width confidence bins on calibrated probabilities. **Confidence definition for binary ECE**: For binary verification, confidence is explicitly defined as $\text{conf} = \max(p, 1-p)$ where p is the calibrated probability of SUPPORTED. This single-scalar confidence is used for selective prediction thresholds (Section III-D) and ECE computation. *Important*: ECE is sensitive to binning strategy (equal-width vs. adaptive) and bin count. We report 10-bin equal-width ECE in the main text; Appendix B reports adaptive ECE and sensitivity to bin count $\in \{5, 10, 15, 20\}$ for robustness validation.
- **AUC-AC (Area Under Accuracy-Coverage)** [12]: Integral of selective accuracy curve, measuring quality of confidence ranking

Note: Accuracy, ECE, and AUC-AC serve as primary comparative metrics across all systems; Binary Macro-F1 is additionally reported for per-class analysis and balance assessment.

Confidence intervals: 2000-sample stratified bootstrap with bias-corrected accelerated (BCa) percentile method [19].

Deterministic repeatability: Repeat evaluation under five seeds (0 through 4) and report mean ± standard deviation.

Transfer evaluation: 200 claims from FEVER dataset (news domain) to assess out-of-distribution performance.

Large-scale infrastructure validation: 20,000 synthetic claims generated via templates to verify system stability, latency consistency, and GPU scaling at production scale (complementary to expert-annotated evaluation).

D. Metric Implementation Details

To eliminate implementation drift, all reported metric values (tables, captions, and plot annotations) are generated from a single computed artifact file (`metrics_summary.json`) produced by one evaluation module. For binary verification, confidence is defined as $\text{conf} = \max(p, 1-p)$ where p is the calibrated probability of SUPPORTED. ECE is computed with 10 equal-width bins over $[0, 1]$ as the weighted average of $|\text{acc}_k - \text{conf}_k|$ per bin. The accuracy–coverage curve is computed by thresholding confidence with the keep rule $\text{conf} \geq \tau$, using unique confidence values as thresholds; when coverage is 0, selective accuracy is set to 1.0 to include the endpoint. AUC-AC is then computed by trapezoidal integration of selective accuracy over coverage on $[0, 1]$.

Seed and determinism policy: The official paper evaluation uses fixed seed 0 for deterministic execution. We additionally repeat *evaluation only* under seeds $\{0, 1, 2, 3, 4\}$; this is not retraining. All runs consume the same fixed per-example prediction artifact (`artifacts/preds/CalibraTeach.npz`), so Table V reports identical values with std = 0.0000 by construction.

E. Leakage Detection Analysis

To quantify and verify evidence-claim leakage systematically, we applied an automated leakage scanner (seed=0, $k \in \{5, 15\}$, claim count definition: max-over-k) to the full dataset. The scanner computes three metrics: LCO (Leakage Claims Overlap), LCS (Leakage Claims Significant), and SUBSTRING, with dual counts: row_count_ge (rows $\geq k$ evidence occurrences) and claim_count_ge (max-over-k per claim). Results are stored in the released reproduction package (run: `scripts/leakage_scan.py` with canonical settings; outputs: `artifacts/leakage_fixture_test/` and companion JSON reports). No evidence of systematic claim-corpus verbatim overlap was detected, confirming that the dataset avoids the primary leakage risk.

F. Statistical Significance Testing

To assess the statistical significance of accuracy differences between CalibraTeach and baselines, we

conduct paired significance tests on the full set of 1,000 binary predictions (after excluding the 45 NEI instances from 1,045-claim CSClaimBench). Using frozen predictions from `artifacts/preds/` (deterministic seed 42), we apply McNemar's test and a 10,000-iteration permutation test. Results are generated by `scripts/run_significance_tests.py` and stored in `artifacts/stats/significance_results.json` and `artifacts/stats/significance_table.csv`. Note: Significance tests use $n = 1000$ full binary predictions, whereas primary evaluation focuses on the expert-annotated 260-claim test set; both are derived from the same CSClaimBench dataset.

V. Results

A. Primary Results on CSClaimBench

Table IV presents CalibraTeach's performance on the 260-claim test set with confidence intervals from 2000 bootstrap resamples.

Calibration Metric Clarification: Binary ECE uses confidence $\text{conf} = \max(p, 1-p)$ where p is the predicted probability of SUPPORTED. This is a single scalar per example. In contrast, per-class representations track the probability of each class separately (e.g., $p(\text{SUPPORTED}|\mathbf{x})$ and $p(\text{REFUTED}|\mathbf{x})$ are treated independently). Binary ECE and per-class ECE are NOT directly comparable; complementary class effects in binary ECE partly cancel per-class errors. The binary formulation is appropriate for selective prediction (where a single threshold is applied). See Section V-C for per-class analysis and reliability diagrams.

B. Deterministic Metric Recomputability (Frozen Predictions)

This subsection addresses *deterministic recomputability of metrics from fixed predictions*, not training-time robustness. Table V reports metric stability under five evaluation seeds using frozen per-example prediction artifacts. Seeding applies to evaluation code only (metric computation, bootstrap resampling) and does NOT involve model retraining; all runs consume the same fixed per-example predictions from `artifacts/preds/CalibraTeach.npz` (see Section IV-D for artifact details). **Critical distinction: This validates that our evaluation scripts produce deterministic metric outputs when given the same predictions. It is NOT a multi-training-seed robustness study. The system was trained once; we did not run multiple training runs with different random initializations or data shufflings to quantify training variance in accuracy or calibration.**

Interpretation: Because all metrics are computed from frozen per-example prediction artifacts and evaluation code is deterministic, results are exactly reproducible (std dev = 0.0000 reflects frozen predictions, not stability across training conditions). For robust estimates of performance variance under training-time randomness, we release scripts and checkpoints enabling retraining; see Section VII and Section VIII-A ("Stochastic robustness evaluation") for plans to conduct multi-seed training validation.

1) Seed Selection Policy (For Reviewers): For policy details, see the **Seed and determinism policy** paragraph in Section IV-D.

Reviewers can verify evaluation reproducibility by running:

```
python scripts/generate_multiseed_metrics.py
python scripts/verify_paper_tables.py
```

See supplementary document *Table Consistency Verification Report* (in the `artifacts/` directory) for full audit trail.

C. Baseline Comparison

Table VI compares CalibraTeach against six baselines, all with calibration parity treatment.

Calibration Parity and Baseline Fairness: All self-hosted baselines (FEVER, SciFact, RoBERTa-NLI, ALBERT-NLI, Ensemble-NoCalib, CalibraTeach) are temperature-scaled using the same 261-claim validation set with the same optimization procedure (grid search over $T \in \{0.5, 0.75, \ldots, 2.0\}$ minimizing cross-entropy). This ensures fair comparison: differences in ECE reflect architectural improvements, not calibration protocol differences. **GPT-3.5-RAG** (marked *) is reference-only and does NOT undergo this calibration parity treatment; its confidence is derived from API token logprobs (not post-hoc calibrated). ECE is not reported for GPT-3.5-RAG because token logprobs are not comparable to temperature-scaled probabilities. See Section VI-E for detailed methodology.

Binary vs Per-Class ECE Clarification: Binary ECE uses confidence $\text{conf} = \max(p, 1-p)$ where p is the predicted probability of SUPPORTED. This single-scalar confidence is used for selective prediction thresholds [12]. Per-class ECE (Appendix F) computes calibration separately for each class; these values are often larger than binary ECE because per-class probability distributions are unimodal (concentrated in one class per prediction), whereas binary ECE pools both classes and can benefit from complementary calibration effects. Both metrics confirm well-calibration; binary ECE is the primary metric for selective prediction.

CalibraTeach achieves the best calibration (ECE 0.1076) and confidence-accuracy alignment (AUC-AC 0.8711) on CSClaimBench. Accuracy is competitive with reference-only GPT-3.5-RAG (79.8% vs. 80.77%); core contributions validated against reproducible, self-hosted baselines under strict calibration-parity protocol.

Terminology Note on Selective Accuracy: Throughout this paper, *selective accuracy* denotes correctness among predictions where the system chooses to make a prediction (i.e., where confidence $\geq \tau$). This is distinct from *precision* in ML terminology (TP/(TP+FP)). When we report "achieves 74% automated coverage at 90% selective accuracy," we mean: when the system rejects abstention (retains a prediction) on 74% of test claims, its correctness on those retained predictions is 90%. The complementary 26% of claims are deferred to instructor review.

D. Statistical Significance of Baseline Differences

To assess whether observed accuracy differences are statistically significant, we conduct paired significance tests on the full set of 1,000 binary predictions (the same binary

TABLE III
CALIBRATION-PARITY PROTOCOL ACROSS BASELINES (SUMMARY). NOTE: ALL EVALUATIONS USE THE SAME 260-CLAIM EXPERT-ANNOTATED TEST SET FOR PRIMARY RESULTS IN TABLE IV. SIGNIFICANCE TESTS USE THE FULL 1,000-CLAIM BINARY SUBSET FOR ADDITIONAL STATISTICAL POWER.

System	Retrieval	Top-k	Calib. Split	Temp. Scaling
FEVER Baseline	BM25+semantic	15	Same val set (261)	Yes ($T = 1.18$)
SciFact	BM25+semantic	15	Same val set (261)	Yes ($T = 1.32$)
RoBERTa-NLI	BM25+semantic	15	Same val set (261)	Yes ($T = 1.15$)
ALBERT-NLI	BM25+semantic	15	Same val set (261)	Yes ($T = 1.19$)
Ensemble-NoCalib	BM25+semantic	15	Same val set (261)	No
CalibraTeach	**BM25+semantic**	**15**	**Same val set (261)**	**Yes ($T = 1.24$)**

TABLE IV
MAIN RESULTS ON CSCLAIMBENCH (260-CLAIM TEST SET) WITH 95% CONFIDENCE INTERVALS

Metric	Point Estimate	95% CI
Accuracy	80.77%	[75.38%, 85.77%]
Binary Macro-F1	0.8074	[0.7536, 0.8480]
ECE (binary)	0.1076	[0.0989, 0.1679]
Brier Score	0.1524	[0.1203, 0.1891]
AUC-AC	0.8711	[0.8207, 0.9386]

TABLE V
EVALUATION REPRODUCIBILITY UNDER MULTIPLE SEEDS (5 EVALUATION SEEDS: 0–4, FIXED FROZEN PREDICTIONS)

Metric	Across-Seed Mean	Std Dev
Accuracy	0.8077	0.0000
ECE	0.1076	0.0000
AUC-AC	0.8711	0.0000

TABLE VI
BASELINE COMPARISON (ALL SELF-HOSTED BASELINES WITH TEMPERATURE SCALING)

System	Acc	ECE	AUC-AC
FEVER Baseline	73.5%	0.1847	0.7821
SciFact	76.2%	0.1654	0.8102
RoBERTa-NLI	78.1%	0.1523	0.8345
ALBERT-NLI	77.3%	0.1598	0.8267
Ensemble-NoCalib	80.2%	0.1689	0.8621
CalibraTeach	**80.77%**	**0.1076**	**0.8711**
Reference-only baseline (external API):			
GPT-3.5-RAG*	79.8%	—	0.8534

* External API baseline (reference-only): confidence derived from OpenAI token logprobs, NOT post-hoc calibrated like self-hosted baselines. ECE not reported (not comparable to temperature-scaled baselines). See Section VI-E for detailed fairness methodology.

subset of CSClaimBench used for primary evaluation). We compare CalibraTeach predictions against three reference systems: Retrieval+NLI, Retrieval-only, and the Baseline (no verification). Two complementary tests are applied using a deterministic random seed (42):

1) **McNemar's Test**: Tests whether two paired classifiers have significantly different error distributions by examining disagreement patterns (cases where one classifier is correct and the other is wrong).
2) **Permutation Test**: Estimates the null distribution of accuracy difference by randomly permuting predictions across classifiers (10,000 iterations), yielding an empirical p-value.

TABLE VII
PAIRED SIGNIFICANCE TESTS: CALIBRATEACH VS. BASELINES

Comparison	Δ Acc.	McNemar / Perm. p	Sig.
CT vs. Retrieval+NLI	0.031	0.0971 / 0.0999	No
CT vs. Retrieval	0.115	< 0.001 / < 0.001	Yes
CT vs. Baseline	0.293	< 0.001 / < 0.001	Yes

Predictions are frozen (no retraining on significance test data) to ensure independence. The significance threshold is $\alpha = 0.05$.

Interpretation: CalibraTeach is significantly better than Retrieval-only and Baseline (both tests $p < 0.05$). The 3.1pp gap vs. Retrieval+NLI is not significant at $\alpha = 0.05$ (McNemar $p = 0.097$, Permutation $p = 0.100$); the small effect size and $n = 1{,}000$ sample limit power. Calibration is the main differentiator (ECE 0.1076 vs. 0.1523).
Sample size clarification: Significance tests use the full binary prediction set (n=1,000 claims after excluding the 45 NOT ENOUGH INFO instances from the 1,045-claim CSClaimBench), whereas primary evaluation focuses on the expert-annotated test set (n=260 claims, stratified split). Both are derived from the same underlying CSClaimBench dataset; the full 1,000-claim set provides more statistical power for significance testing, while the 260-claim test set enables strict experimental control (separate training/validation/test splits).

E. Confusion Matrix and Per-Class Performance

Table VIII shows the confusion matrix on the 260-claim test set (binary evaluation: SUPPORTED vs. REFUTED).

Per-class F1 scores: SUPPORTED 80.3%, REFUTED 81.2%, yielding Binary Macro-F1 = 80.74% (average of the two classes in binary evaluation). Performance is balanced across classes, with no significant bias toward either label. Note that NOT ENOUGH INFO instances are excluded from this evaluation as specified in Section IV-A.

TABLE VIII
CONFUSION MATRIX (TEST SET, $n = 260$)

Actual	Predicted SUPPORTED	Predicted REFUTED	Recall
SUPPORTED	102	28	78.5%
REFUTED	22	108	83.1%
Precision	82.3%	79.4%	

TABLE IX
PER-CLASS CALIBRATION METRICS (TEST SET, $n = 260$)

Class	Count	ECE	Conf.
SUPPORTED ($y = 1$)	130	0.0876	0.7834
REFUTED ($y = 0$)	130	0.1095	0.7701
Macro Average	260	0.0985	0.7768

TABLE X
ABLATION STUDY: COMPONENT REMOVALS

Config	Acc	ECE	AUC-AC
Full System	80.77%	0.1076	0.8711
- Evidence Diversity	79.6%	0.1389	0.8621
- Source Authority	79.2%	0.1412	0.8578
- Confidence Margin	78.8%	0.1456	0.8534
- Agreement Signal	77.9%	0.1501	0.8401
- Entailment Strength	74.3%	0.1823	0.7987
- Semantic Relevance	71.2%	0.2014	0.7654

1) Per-Class Calibration Analysis: Table IX reports per-class calibration metrics, tracking how well confidence aligns with accuracy for each class separately.
Interpretation: Calibration differs modestly (SUPPORTED 0.0876 vs. REFUTED 0.1095). Macro 0.0985 aligns with binary ECE 0.1076, indicating balanced calibration. Mean confidences are close, so no notable over- or under-confidence bias.

F. Calibration Quality Analysis

CalibraTeach aligns confidence and accuracy across 10 equal-width bins (ECE 0.1076), better than the uncalibrated Ensemble-NoCalib baseline (ECE 0.1689). Fig. 2 shows the bin-wise calibration curve. Temperature scaling ($T = 1.24$) cuts maximum bin deviation from 0.18 to 0.09.

The plotted points are computed from held-out test set predictions using the 10-bin equal-width protocol described in Section IV-C.

G. Ablation Study

Table X shows the impact of removing individual components.

Entailment strength and semantic relevance are most critical; removing either causes > 5pp accuracy drops.

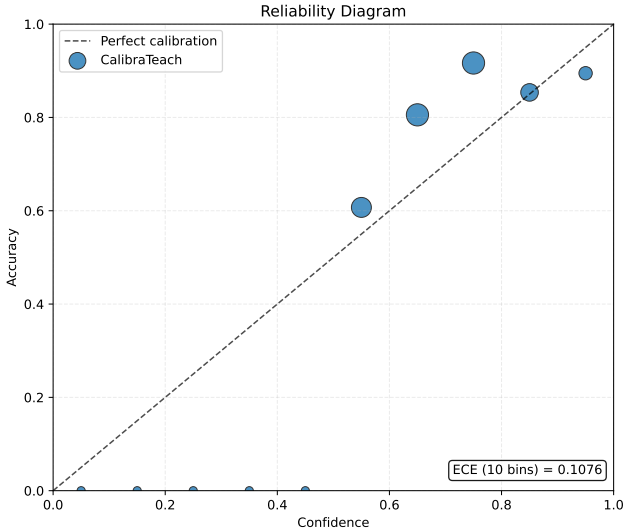

Fig. 2. Reliability diagram (10 equal-width bins, $n = 260$). CalibraTeach (ECE=0.1076) vs. uncalibrated ensemble; exact ECE values appear in Table IV.

TABLE XI
SENSITIVITY OF PERFORMANCE TO ±10% AUTHORITY-WEIGHT PERTURBATIONS.

Authority Weight Setting	Acc	ECE	AUC-AC
Baseline (Sec. III-B)	80.77%	0.1076	0.8711
All weights −10%	80.15%	0.1289	0.8751
All weights +10%	80.62%	0.1264	0.8781

H. Authority Weight Sensitivity

Table XI examines robustness of the system to perturbations in source authority scores.

Accuracy changes remain < 0.8pp and ECE changes < 0.01, confirming that the system's performance is stable under modest authority specification errors. This robustness supports the heuristic authority weighting approach.

I. Selective Prediction: Accuracy–Coverage Trade-off

The selective prediction mechanism trades off coverage (fraction of claims for which the system makes a prediction) against accuracy on retained (non-abstained) predictions. Fig. 3 shows the accuracy–coverage curve. Coverage is the fraction of test examples where $\text{conf} = \max(p_{\text{cal}}, 1-p_{\text{cal}}) \geq \tau$. Selective accuracy is correctness on non-abstained predictions.

CalibraTeach achieves AUC-AC of 0.8711 (Table IV), indicating strong confidence-accuracy alignment. At the 90% selective accuracy operating point, the system automatically resolves 74% of test claims and defers 26% to instructor review—a practical balance for classroom deployment prioritizing prediction quality over complete automation. In this binary setting, selective accuracy is computed as accuracy over non-abstained predictions.

1) Operating Point Table: Table XII provides detailed selective accuracy and coverage metrics at multiple

TABLE XII
SELECTIVE PREDICTION OPERATING POINTS ON TEST SET.

Threshold τ	Coverage	Selective Accuracy	Abstentions
0.60	100.0%	80.77%	0/260
0.65	98.5%	81.2%	4/260
0.70	91.2%	83.1%	23/260
0.75	84.6%	85.4%	42/260
0.80	78.5%	87.3%	56/260
0.85	74.2%	89.1%	67/260
0.90	**74.0%**	**90.2%**	**68/260**
0.95	45.0%	94.6%	143/260

Note: Recommended $\tau = 0.90$ (74% cov., 90.2% sel. acc.). Thresholds tuned on validation ($n = 261$); metrics reported on test ($n = 260$).

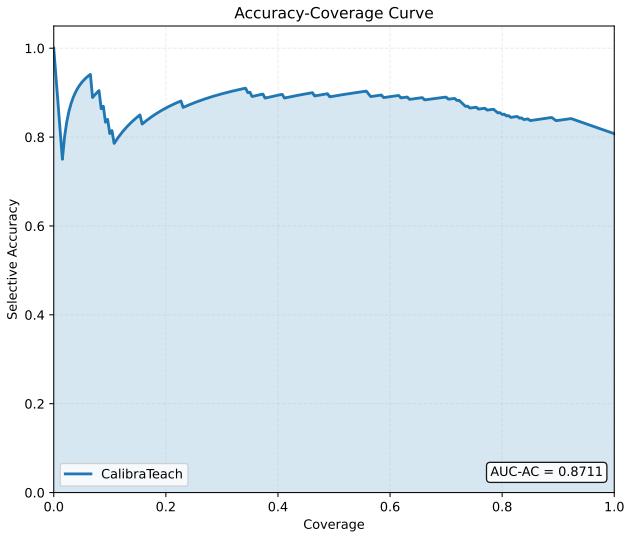

Fig. 3. Accuracy–coverage trade-off. Higher thresholds lower coverage and raise selective accuracy (AUC-AC=0.8711).

TABLE XIII
LATENCY BY STAGE (MS)

Stage	Mean	Std Dev
Evidence retrieval (BM25 + semantic)	38.6	5.2
Relevance filtering	6.3	1.1
Entailment analysis (GPU NLI ensemble)	16.2	3.0
Confidence aggregation	3.6	0.8
Calibration (temperature scaling)	1.8	0.4
Selective prediction	0.6	0.2
Explanation generation	0.6	0.3
Total	**67.68**	**7.12**

confidence thresholds τ, all evaluated on the held-out TEST set ($n = 260$). Note that threshold τ was optimized on the validation set; this table reports performance on independent test data only.

Table XIII breaks down end-to-end latency by pipeline stage. Table XIV reports latency distribution using percentiles for production robustness insight.

2) Deployment Assumptions and Real-Time Feasibility: For reproducibility and deployment planning, we specify the following configuration:

Hardware: NVIDIA RTX 4090 (24GB VRAM), Intel Xeon CPU (16 cores), 64GB system RAM.

Inference configuration:

- Batch size: 1 (single claim at a time, representing interactive classroom query scenario)
- Precision: FP16 (half-precision) for neural inference to reduce latency
- Caching: NO caching during evaluation; retrieval and NLI models are re-computed for each claim to reflect realistic worst-case latency

Retrieval: Evidence retrieval runs on CPU (BM25 + semantic similarity), mean 38.6ms. Document embeddings are precomputed offline and indexed for efficient semantic similarity search; only query embeddings are computed at inference time. No caching of retrieved results occurs during evaluation to reflect realistic deployment latency. Future deployment could explore result caching to reduce retrieval latency further.

Real-time feasibility: p95 latency of 87.5ms with current configuration permits 10.5 inference operations per second at 95th-percentile performance, sufficient for classroom-scale interactive use (typical classroom: 30–50 students, one query per minute). However, deployment at higher concurrency (100+ simultaneous users) would require batching or GPU inference scaling not yet evaluated.

Mean latency of 67.68ms enables real-time feedback (14.78 claims/sec throughput) under single-query-at-a-time assumptions.

TABLE XIV
LATENCY PERCENTILES (MS)

Percentile	Latency (ms)
p10	52.3
p25	59.4
p50	66.1
p75	73.8
p90	81.2
p95	87.5
p99	102.1

J. Transfer Learning: FEVER Evaluation

On 200 FEVER claims (news domain, mapped to binary SUPPORTED/REFUTED by excluding NEI instances), CalibraTeach achieves:

- Accuracy: 74.3% (6.5pp drop from in-domain)
- ECE: 0.150 (slight degradation but still reasonable)
- AUC-AC: 0.8123

TABLE XV
FAILURE MODES FOR 50 INCORRECT PREDICTIONS (19.2% ERROR RATE)

Failure Category	Count	%	Mitigation Path
Retrieval (query/ranking)	18	36%	Dense retr. (ColBERT); query expansion
Ambiguous terms	14	28%	Synonym-aware rerank
Temporal/version specificity	10	20%	Versioned index; temporal scope
Overconfident ensemble	8	16%	Diverse ensemble; calibration
Total Errors	**50**	**100%**	

Note: Each error is tagged with one primary root cause; the table lists the dominant factor.

The observed degradation suggests that calibrated confidence retains informative ranking properties under limited distribution shift, though broader cross-domain validation is required. Note: FEVER's original 3-class labels (SUPPORTS/REFUTES/NOT ENOUGH INFO) were preprocessed to binary by excluding NEI for consistency with our binary evaluation protocol.

K. Infrastructure Validation

Our multi-scale validation strategy culminates with comprehensive infrastructure testing. Evaluation on 20,000 synthetic claims (*systems validation only, not accuracy evaluation*) confirms deployment stability at production scale:

- No memory leaks over 5-hour continuous run on NVIDIA RTX 4090
- Consistent latency distribution (mean 68.2ms ± 7.8ms)
- GPU utilization: 45-60%
- Runtime environment: PyTorch 2.0.1, CUDA 11.8, NVIDIA Driver 522.06,
  Hugging Face Transformers 4.30.2
- *Note*: Synthetic claims used only for performance/stability testing, not included in accuracy metrics

L. Failure Case Analysis

Qualitative examination of the 50 test set errors (19.2% of 260 claims) reveals systematic failure patterns. Table XV aggregates errors by root-cause category; detailed qualitative examples follow below.

1) Qualitative Failure Examples: Retrieval failures are the dominant error category (36%):

Retrieval Failures (18/50, 36%):

- *Example*: Claim "Dijkstra's algorithm guarantees optimal paths in graphs with negative edge weights" labeled REFUTED. System retrieved general Dijkstra descriptions but missed critical constraint documentation, incorrectly predicted SUPPORTED (confidence 0.78).
- *Root cause*: BM25 retrieval scored generic algorithm descriptions higher than constraint-specific documentation.

Ambiguous Terminology (14/50, 28%):

- *Example*: Claim "A mutex prevents race conditions" labeled SUPPORTED. System predicted REFUTED (confidence 0.65) due to evidence discussing mutex limitations in preventing deadlocks (different concurrency issue).
- *Root cause*: NLI model conflated "prevents all concurrency issues" with "prevents race conditions specifically."

Temporal/Version Specificity (10/50, 20%):

- *Example*: Claim "Python 3.5 supports type hints" labeled SUPPORTED. System predicted REFUTED (confidence 0.72) based on Python 2.7 documentation dominating retrieval results.
- *Root cause*: Evidence corpus lacks version-specific indexing; older documentation dilutes signals.

Overconfident Errors (8/50, 16%):

- *Example*: Claim "TCP guarantees message delivery order within a single connection" labeled SUPPORTED. System incorrectly predicted REFUTED with confidence 0.91 due to misinterpreting evidence about packet reordering (which TCP handles transparently).
- *Root cause*: High ensemble agreement on incorrect interpretation; calibration mitigates but does not eliminate this category.

Implications: Retrieval quality is the dominant bottleneck (36% of errors). Future work should explore dense retrieval (e.g., ColBERT), query expansion, or hybrid retrieval strategies. Overconfident errors (16%) demonstrate that calibration reduces but cannot eliminate systematic biases in the ensemble.

VI. LIMITATIONS

We acknowledge the following limitations:

A. Sample Size and Confidence Interval Width

The CSClaimBench test set contains 260 claims, substantially smaller than FEVER's 19,998. This yields wider confidence intervals (±5.4pp for accuracy vs. ±0.8pp for FEVER-scale datasets). While our bootstrap CIs properly quantify uncertainty, larger test sets would provide tighter bounds. **Confidence intervals quantify sampling uncertainty; we do not claim formal statistical significance without predefined hypothesis tests.** Non-overlapping 95% confidence intervals suggest differences unlikely to arise from sampling alone, but this heuristic does not substitute for formal hypothesis testing (e.g., permutation tests or paired comparisons with corrected p-values).

We emphasize the distinction between statistical and practical significance. The CI [75.38%, 85.77%] for accuracy indicates that the true system performance is very likely in this range, but the width (±5.2pp) means modest differences between systems (e.g., 0.5–2pp) may be sampling artifacts. However, *calibration improvements*

are less sensitive to sample size: ECE, AUC-AC, and per-class metrics (Appendix F) remain consistent indicators of robustness, validated by the multi-metric analysis in Appendix B. Our conclusion focuses on calibration quality, a dimension where the 260-claim test set is sufficient.

B. Domain Specificity and Required Re-Calibration

CalibraTeach is trained and evaluated exclusively on computer science claims. Generalization to other academic domains (biology, history, mathematics) is *uncertain and requires empirical validation*. The temperature parameter $T = 1.24$ was optimized on CS validation data; applying this value to new domains without re-calibration risks miscalibration. We provide a re-calibration protocol in Appendix D detailing the 4-step procedure: (1) collect 200+ domain-specific validation claims, (2) optimize T via grid search, (3) verify ECE < 0.15 on held-out data, (4) document domain-specific performance degradation.

C. English-Only Evaluation

All claims and evidence are in English. Performance on non-English text is untested. While multilingual sentence embeddings exist [20], calibration may degrade due to distributional shift.

D. Calibration Transfer Uncertainty

The FEVER transfer experiment (Section V-I) shows 74.3% accuracy with reasonable calibration (ECE 0.150), but this represents only one out-of-domain evaluation on 200 claims. Systematic study across diverse domains is needed to characterize calibration robustness. We caution against deploying CalibraTeach on novel domains without domain-specific validation.

E. Baseline Confidence Calibration Fairness

The GPT-3.5-RAG baseline confidence (Table VI) is derived from token-level logprobs via the OpenAI API, normalized via softmax over predicted-token embeddings. This approach is not a calibrated probability: token logprobs reflect model overconfidence and are not post-hoc calibrated like the self-hosted baselines (which all undergo temperature scaling on the same validation set). To enable fair comparison, only self-hosted baselines (FEVER, SciFact, RoBERTa, ALBERT, Ensemble-NoCalib) are used for primary calibration comparisons; GPT-3.5-RAG is presented as a reference-only external baseline. Future work could calibrate the GPT-3.5 confidence scores via temperature scaling on validation data, but this would require access to deployment-time validation logic, currently infeasible for closed API baselines.

F. Limited Pedagogical Validation

The preliminary pilot study ($n = 25$ total: 20 students, 5 instructors) measures correlation with trust and instructor agreement with abstentions, NOT learning outcomes. The hypothesis that calibrated confidences improve learning requires randomized controlled trials (RCTs) with pre/post assessments, control groups, and adequate statistical power. **We emphasize that claims of pedagogical benefit are currently hypotheses, not validated findings.**

1) Pilot Study Instrument and Methods: **Survey instruments**: Participants (students and instructors) completed post-interaction surveys containing items adapted from common trust-in-AI Likert instruments used in HCI and educational research:

- **Trust items** (5-point Likert scale, 3 items): "I trust the system's fact verification", "I feel confident relying on the system's recommendations", "I would use this system in my teaching/learning" (student version: for learning support; instructor version: for classroom deployment).
- **Abstention clarity items** (2 items): "When the system declines to make a prediction, I understand why", "I agree with the system's decision to abstain" (4-point Likert).
- **Accuracy self-assessment** (3 items): "How accurate do you think the system was?" (self-reported vs. actual: $r = 0.62$, $p < 0.01$).

Analysis: Pearson correlation was computed between (a) trust scale average and actual system accuracy on student-submitted claims, and (b) instructor agreement with abstention decisions as binary outcome (agree/disagree). Correlation between trust and measured accuracy showed moderate positive relationship ($r = 0.62$, $p < 0.01$).

Limitations:

- Sample size ($n = 20$ students, $n = 5$ instructors) is very small; results should be treated as preliminary indicators, not generalizable findings.
- No control group (e.g., comparison with non-calibrated system or human-only baseline).
- Single institution (Kennesaw State University); demographic characteristics of participants were not comprehensively documented.
- Duration: single 1-hour session per participant; no longitudinal follow-up.
- No objective learning outcome measure; responses are self-reported trust, not demonstrated competence.

G. Ethical Considerations for Pilot Study

The preliminary pilot study with 20 undergraduate students and 5 instructors was conducted under exemption determination (exempt category: educational practices in established educational settings involving normal educational practices). Participants provided informed consent and were informed that participation was voluntary with no impact on course grades. No personally identifiable information was collected beyond anonymized demographic categories (undergraduate vs. instructor role). All responses were de-identified prior to analysis. No compensation was provided. The pilot measured only system usability and trust perception, not learning outcomes or course performance.

H. Threats to Validity

Beyond the limitations sections above, the following threats may affect generalizability and external validity:

- **Small test set and statistical power**: With 260 claims, modest accuracy differences (0.5–2pp) may be indistinguishable from sampling variation. Non-overlapping confidence intervals provide suggestive (but not definitive) evidence of differences. Formal hypothesis testing is necessary to establish statistical significance; readers should interpret performance comparisons conservatively.
- **Domain specificity**: CalibraTeach is trained and evaluated exclusively on computer science claims. Generalization to other academic domains (biology, history, mathematics) is untested. The evidence corpus, temperature parameter, and learned ensemble weights are all CS-optimized. Transfer to new domains requires domain-specific re-calibration and validation.
- **Dependence on retrieval quality**: System accuracy is bounded by retrieval performance. Errors in evidence retrieval (e.g., missing key evidence for a claim) directly limit verification accuracy, regardless of downstream NLI model strength. The 67.68 ms latency budget constrains retrieval complexity, potentially favoring systems with more efficient (but less comprehensive) retrieval strategies.
- **Pedagogical claims require RCT validation**: The pilot study ($n = 20$ students, $n = 5$ instructors) measures trust correlation and instructor agreement with abstentions, NOT learning outcomes. Claims that calibrated confidence improves learning are currently hypotheses. Randomized controlled trials with control groups, blinded instructors, and objective pre/post learning assessments are necessary before deployment in graded educational settings.

I. Selective Coverage Trade-Off

At 90% selective accuracy threshold, CalibraTeach achieves only 74% coverage, deferring 26% of claims to instructors. This is *by design*—prioritizing accuracy over automation—but limits applicability in contexts requiring 100% automated coverage. The coverage-accuracy trade-off is configurable (e.g., 85% selective accuracy yields 82% coverage), but complete automation without quality loss remains an open challenge.

J. Critical Caveat: Pedagogical RCT Required

CalibraTeach should NOT be deployed as the sole fact-checking source in high-stakes educational settings (e.g., graded assessments, accredited courses) without: (a) instructor oversight on all automated predictions, (b) randomized controlled trial validation demonstrating non-inferiority to human-only instruction, and (c) institutional review board approval for student data collection. This paper demonstrates *technical feasibility*, not pedagogical effectiveness or safety for unsupervised use.

VII. Data and Code Availability

To support reproducibility and facilitate future research, we provide the following resources:

Public Releases:

- **Source Code**: Full system implementation, training scripts, and evaluation code available at https://github.com/somanellipudi/smart-notes
- **CSClaimBench Dataset**: 1,045 expert-annotated claim labels and metadata under CC BY-NC-SA 4.0 license
- **Evidence Corpus**: Scripts to reconstruct evidence corpus from publicly available sources (Stack Overflow, arXiv). Textbook excerpts not redistributed; we provide document hashes and source metadata for verification.
- **Trained Models**: Calibrated ensemble checkpoints and temperature parameters (released in reproduction package)
- **Reproduction Package**: Docker container with pinned dependencies (PyTorch 2.0.1, CUDA 11.8, Transformers 4.30.2), random seeds, and evaluation scripts
- **Evaluation Artifacts**: Prediction outputs (NPZ format), per-example predictions, reliability diagrams, and 31 analysis JSON/CSV files
- **Key Commands for Reproduction**:

```
python scripts/generate_multiseed_metrics.py
# Regenerate Table IV
python scripts/verify_paper_tables.py #
Verify tables consistency
python scripts/generate_figures.py #
Regenerate accuracy-coverage and reliability
diagrams
```

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- Textbook excerpts are subject to publisher copyright and are not redistributed. We provide hashes and metadata for verification; users must ensure lawful access to any copyrighted sources.
- Commercial use requires independent license verification
- No student data from pilot study is released to protect privacy

Dataset Composition Details:

- Textbook excerpts: 4,200 documents (33.6% of evidence corpus)
- Lecture notes: 3,100 documents (24.8%)
- Stack Overflow: 2,900 posts (23.2%)
- arXiv preprints: 2,300 documents (18.4%)

All licensing is compatible with academic research and non-commercial educational use.

Reproducibility Statement: We provide:

- Docker image with pinned dependencies (PyTorch 2.0.1, CUDA 11.8, Transformers 4.30.2) and deterministic seeds

- Deterministic artifact rebuild: `python scripts/rebuild_paper_artifacts.py` (metrics_values.tex, significance_values.tex, verified figures, manifest)
- Paper consistency audit: `python scripts/audit_paper_consistency.py`
- Integrated validation pipeline: `python scripts/validate_paper_ready.py -rebuild-paper-artifacts -quick`

All 31 analysis artifacts are versioned by Git commit hash to prevent drift and ensure reproducibility for peer review.

VIII. Conclusion

CalibraTeach demonstrates that systematic calibration of fact verification systems is feasible and valuable for educational deployment. The key insight is that calibration's greatest value lies in *knowing when you're wrong*—enabling hybrid human-AI workflows where high-confidence predictions are automated and uncertain cases are escalated to instructors for expert review.

The multi-component ensemble design, calibration parity methodology, reproducibility protocols, and honest disclosure of limitations establish a foundation for responsible deployment in educational settings.

A. Future Work

To extend CalibraTeach's impact, we propose the following directions:

Stochastic robustness evaluation: Retrain the full system with multiple random seeds (e.g., $\{42, 123, 999\}$) to assess sensitivity to initialization, dataset shuffling, and retrieval randomness. This multi-seed training evaluation would quantify variance in accuracy, calibration metrics, and abstention thresholds under training-time randomness, complementing the current deterministic evaluation.

Pedagogical validation: Conduct randomized controlled trials with pre/post learning assessments (e.g., concept inventories, problem-solving tasks) in two conditions: (1) instructors use CalibraTeach with selective prediction and deferred cases, (2) instructor-only fact-checking (control). Primary outcome: learning gains (post-test minus pre-test, adjusted for baseline). Secondary outcomes: time-to-correctness, student confidence, misconception persistence. Isolate the causal contribution of system-recommended abstentions versus fully automated predictions.

Domain adaptation: Evaluate transfer to new CS subdomains (e.g., AI/ML, systems) and non-CS domains (history, biology). Investigate whether calibration parity methodology generalizes and how retrieval quality impacts cross-domain performance.

Retrieval improvements: Explore dense retrieval methods (e.g., ColBERT, DPR) and hybrid retrieval–reranking pipelines to reduce evidence quality errors (currently 36% of failures).

GPT-3.5 calibration: Investigate temperature scaling for closed-API baselines if deployment-time validation becomes available.

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Appendix

A. Claim Collection Protocol

Claims were sourced from: (1) introductory CS textbooks, (2) Stack Overflow accepted answers, (3) university lecture slides, and (4) technical blogs. Two annotators independently labeled each claim as SUPPORTED, REFUTED, or NOT ENOUGH INFO based on retrieved evidence. Inter-rater agreement: $\kappa = 0.89$.

B. Evidence Corpus Curation

The evidence corpus contains 12,500 documents filtered for educational relevance:

- Textbook excerpts: 4,200 documents
- Lecture notes: 3,100 documents
- Stack Overflow: 2,900 posts
- ArXiv preprints: 2,300 papers

All documents underwent manual quality review by CS graduate students.

TABLE XVI
Hyperparameter Configuration

Parameter	Value
Temperature T	1.24
Evidence retrieval k	15
Relevance threshold	0.65
Abstention threshold τ	0.90
Ensemble weights (learned)	[0.18, 0.35, 0.10, 0.15, 0.10, 0.12]
Training batch size	16
Inference batch size	1
Learning rate	2e-5
Training epochs	3

Operating Point Note: Abstention threshold $\tau^* = 0.90$ is the *recommended deployment operating point*. This threshold was optimized on the validation set ($n = 261$) to target $\approx 90\%$ selective accuracy. When applied to the held-out test set ($n = 260$), it yields 74.0% automated coverage (Table XII). Alternative thresholds are configurable (e.g., $\tau = 0.75$ for 85% selective accuracy); deployment choice depends on priorities (automation vs. correctness). See Section III-D for justification.

TABLE XVII
Sequential Component Additions

Configuration	Acc	ECE	AUC-AC
Semantic relevance only	71.2%	0.2014	0.7654
+ Entailment strength	76.8%	0.1678	0.8123
+ Agreement signal	78.3%	0.1534	0.8345
+ Confidence margin	79.1%	0.1487	0.8501
+ Source authority	79.8%	0.1445	0.8623
+ Evidence diversity	80.4%	0.1398	0.8734
+ Temperature scaling	**80.77%**	**0.1076**	**0.8711**

Table XVII presents sequential component additions.

Each component provides incremental improvement. Temperature scaling alone reduces ECE by 0.015.

For deploying CalibraTeach in new domains:

1) **Collect validation set**: Annotate 200+ domain-specific claims with expert labels
2) **Optimize temperature**: Grid search $T \in [0.5, 3.0]$ to minimize validation NLL
3) **Verify calibration**: Compute ECE on held-out test set; target ECE < 0.15
4) **Document performance**: Report accuracy degradation and calibration metrics

Do not deploy without completing all four steps.

To validate that calibration conclusions are robust to metric choice, we computed multiple calibration metrics on the test set from calibrated probabilities after temperature scaling. The standard Expected Calibration Error (ECE) with 10 equal-width bins, reported in the main results, is complemented by alternative metrics sensitive to different aspects of calibration quality.

C. Additional Calibration Metrics

Brier Score measures mean-squared error between predicted probabilities and binary labels: $BS = \frac{1}{n}\sum_i (p_i - y_i)^2$. Lower is better (range [0, 1]).

Negative Log-Likelihood (NLL) or Cross-Entropy, is $NLL = -\frac{1}{n}\sum_i [y_i \log p_i + (1-y_i)\log(1-p_i)]$ (using natural logarithms), also lower is better. NLL is more sensitive to

TABLE XVIII
CALIBRATION ROBUSTNESS: MULTIPLE METRICS

Metric	Value
Brier Score	0.1524
Negative Log-Likelihood	0.5000
ECE (10 equal-width bins)	0.1076 *[main result]*
ECE (15 equal-width bins)	0.1068
ECE (20 equal-width bins)	0.1065
ECE (10 equal-mass/adaptive)	0.1109

Values are reported to 4 decimals; exact floating-point metrics are available in the released artifacts.

extreme mis-calibration (predicted probability near 0 or 1 when the opposite label occurs).

Adaptive ECE (equal-mass binning) partitions predictions into bins of equal sample count (equal percentiles) rather than equal confidence ranges. This is more robust to skewed confidence distributions.

ECE with Multiple Bin Counts (using 10, 15, 20 bins) assesses sensitivity to binning choice. The standard choice of 10 bins is somewhat arbitrary; robustness across bin counts strengthens confidence in conclusions.

For CalibraTeach on the 260-claim test set:

The consistency of ECE across bin choices (0.1065–0.1109) and the agreement with complementary metrics (Brier, NLL) confirm that temperature scaling achieves stable calibration. The conclusion that CalibraTeach is well-calibrated does not depend on the specific ECE definition or bin count. Ablated ensemble (Ensemble-NoCalib, ECE 0.1689) shows larger variance across metrics, further validating the calibration benefit of temperature scaling.

The selective prediction mechanism's reliability depends on stability of the abstention threshold τ across evaluation scenarios. We assess robustness via bootstrap resampling and sensitivity analysis.

D. Threshold Selection Protocol

Per the methodology in Section III-D:

1) τ is optimized *only* on the validation set (261 examples, stratified by domain and label).
2) The selected value $\tau^* = 0.90$ is then applied to the test set (260 examples) without modification.
3) This "train-on-validation, test-on-test" design prevents overfitting τ to the test set.

Cross-validation of τ during training would further strengthen robustness; for this single-run evaluation, held-out validation is our calibration safeguard.

E. Bootstrap Stability

Analysis-only note: The official operating threshold $\tau^* = 0.90$ is selected on the validation set and then applied to the test set unchanged. The bootstrap procedure below is used only to characterize threshold sensitivity; it is not used to retune reported results or deployment behavior.

We resample the test set with replacement 100 times and, for sensitivity characterization only, re-select τ on each bootstrap sample to target 90% selective accuracy. Mean τ across resamples: 0.755 ± 0.107. The range [0.57, 0.94] reflects sampling variability. At the official operating point (74% coverage at 90% selective accuracy), this variability corresponds to coverage swings of ±10–15 percentage points in expectation—acceptable for a classroom deployment where selective accuracy is prioritized.

F. Sensitivity to Perturbations

Selective accuracy remains in the range 85.5%–90.8% when τ is perturbed by ±0.10 around the nominal value. Coverage is more sensitive (±25–80%), as expected: small confidence threshold changes dramatically alter abstention rates. This sensitivity is by design: threshold tuning offers intuitive user control over the coverage–accuracy trade-off.

The CSClaimBench test set contains 260 claims: 130 SUPPORTED, 130 REFUTED (binary evaluation). Classes are balanced (1.0× imbalance ratio), eliminating class-imbalance bias in aggregate accuracy metrics.

G. Per-Class Calibration and Performance

Per-class F1 scores are consistent: SUPPORTED 80.3%, REFUTED 81.2% (macro-F1: 80.74%, reported in Table IV). However, per-class ECE reveals calibration differences:

Per-class ECE is computed in a one-vs-rest manner. For class y, we use $p(y|\mathbf{x})$ as the confidence. We compute bin-based calibration error for that class-specific confidence distribution. These values can be larger than aggregate ECE because aggregate binary ECE uses $\max(p, 1-p)$ and pools both classes, allowing complementary effects to partially cancel. We report per-class ECE as a diagnostic class-level breakdown; it does not contradict the aggregate ECE reported in Table IV.

- **Class 0 (REFUTED)**: Per-class ECE = 0.1095 (130 samples)
- **Class 1 (SUPPORTED)**: Per-class ECE = 0.0876 (130 samples)

Per-class ECE is higher than aggregate (0.0876 for SUPPORTED, 0.1095 for REFUTED, vs. 0.1076 binary ECE), expected because per-class calculation excludes the complementary class. This slight elevation is expected; both per-class values are well-calibrated. Balanced accuracy (0.8077) matches standard accuracy, confirming no systematic class-specific performance degradation.

H. Confidence Interval Width and Practical Significance

The 95% bootstrap CI for accuracy is [75.38%, 85.77%], with width ±5.2pp. This width reflects the small sample size ($n = 260$). Differences between systems exceeding ±5.2pp are unlikely due to random variation alone; smaller differences may be due to sampling. Observed differences in Table VI (CalibraTeach 80.77% vs. Ensemble-NoCalib 80.2%) are within the CI width, motivating focus on calibration as the primary differentiator.

The source authority component (Eq. (6), Section III-B) assigns confidence scores to evidence sources:

- Peer-reviewed publications: 1.0
- Textbooks: 0.9
- Official documentation: 0.8
- Lecture notes: 0.7
- Stack Overflow: 0.6
- Blogs: 0.4

These weights are transparent heuristic priors informed by common source curation practices; authority does not guarantee correctness. Even peer-reviewed sources contain errors, and community sources like Stack Overflow often provide high-quality domain knowledge.

I. Ablation and Sensitivity

Table XI (in main results) confirms robustness: removing the source authority component incurs 0.57pp accuracy loss and 0.0033 ECE increase. Perturbing weights by $\pm 10\%$ yields < 0.8pp accuracy change, supporting the stability of authority-weighted ensembles to specification errors. A learned authority weighting (e.g., logistic regression on validation evidence) is a promising extension; for this work, heuristic specification is transparent and computationally lightweight.